01 평면좌표

9~20쪽

A단계 0001 5 0002 8 0003 4 0004 $\sqrt{10}$ 0005 $\sqrt{26}$

0006 5 0007 $\frac{10}{3}$ 0008 $\frac{10}{7}$ 0009 2 0010 $\left(\frac{11}{3}, -\frac{10}{3}\right)$

0011 $\left(\frac{17}{4}, -1\right)$ 0012 $(4, -2)$

0013 (가) $\frac{y_2+y_3}{2}$ (나) 1 (다) $\frac{x_1+x_2+x_3}{3}$ (라) $\frac{y_1+y_2+y_3}{3}$

0014 $\left(\frac{4}{3}, -1\right)$ 0015 $(1, 2)$

B단계 0016 ② 0017 5 0018 13 0019 5 0020 8

0021 ① 0022 ⑤ 0023 ② 0024 ⑤ 0025 $\sqrt{5}$ km

0026 ② 0027 3 0028 $4\sqrt{10}$ 0029 44 0030 ① 0031 $(1, 3)$

0032 ③ 0033 ⑤ 0034 $(2\sqrt{3}, -\sqrt{3})$

0035 (1) $\angle A=90°$인 직각이등변삼각형 (2) $\frac{1}{2}(a^2+b^2)$

0036 (가) M (나) $(-c, 0)$ (다) $a^2+b^2+c^2$ (라) c^2 0037 풀이 5쪽

0038 ② 0039 풀이 5쪽 0040 $\left(1, \frac{5}{2}\right)$ 0041 4

0042 C, B, A 0043 $(8, -2)$ 0044 ① 0045 ①

0046 $\frac{1}{3}<a<\frac{5}{7}$ 0047 2 0048 10 0049 ③

0050 $(-3, 6)$, $(1, 2)$ 0051 ⑤ 0052 ③ 0053 -2 0054 ③

0055 $(8, -4)$ 0056 ③ 0057 풀이 8쪽 0058 $(2, 8)$

0059 C(14, 0), D(18, 8) 0060 1 0061 $(9, -1)$

0062 ② 0063 $\left(\frac{5}{2}, -\frac{1}{2}\right)$ 0064 ③ 0065 ⑤ 0066 $6+3\sqrt{2}$

0067 $6x-3y-4=0$ 0068 $2x+y-7=0$ 0069 $2x-3y-18=0$

C단계 0070 116 0071 ② 0072 $20\sqrt{2}$ 0073 ② 0074 $\frac{16}{9}$

0075 ⑤ 0076 28 0077 ① 0078 24 0079 $(4, 5)$ 0080 36

0081 7 0082 67 0083 ① 0084 ①

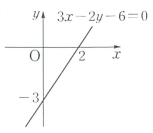
02 직선의 방정식

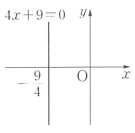
23~40쪽

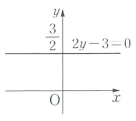
A단계 0085 $y=-2x+4$ 0086 $x=-9$ 0087 $y=2$

0088 $y=-2x+3$ 0089 $y=x-5$ 0090 $x=3$

0091 (가) a (나) b (다) a (라) $\frac{b}{a}$ 0092 $\frac{x}{3}-\frac{y}{6}=1$

0093 

0094 

0095 

0096 제1사분면 0097 제2사분면 0098 $(2, -3)$

0099 $(-1, 1)$ 0100 $4x-5y=0$ 0101 $x=-1$

0102 $-\frac{3}{2}$ 0103 $-\frac{1}{4}$ 0104 평행한 직선: ㄴ, 수직인 직선: ㄷ 0105 -3

0106 -6 0107 $y=-3x-18$ 0108 $y=\frac{2}{5}x-6$

0109 $y=-\frac{1}{2}x-2$ 0110 0 0111 1 0112 $\frac{9}{5}$ 0113 $3\sqrt{5}$

0114 $2\sqrt{10}$ 0115 $2\sqrt{2}$ 0116 9 0117 3 0118 $\sqrt{13}$ 0119 $\frac{11}{10}$

B단계 0120 $y=2x+2$ 0121 -12 0122 $y=1$ 0123 ①

0124 ③ 0125 ⑤ 0126 ④ 0127 $y=-3x+14$ 0128 13

0129 ③ 0130 ③ 0131 $\frac{x}{2}-\frac{y}{8}=1$ 0132 $\frac{x}{8}+\frac{y}{4}=1$

0133 $\sqrt{3}$ 0134 ⑤ 0135 $y=2x+1$ 0136 ④ 0137 ②

0138 $\frac{3}{2}$ 0139 1 0140 $y=-6x+9$ 0141 ①

0142 제1사분면 0143 ㄴ, ㄹ 0144 ③ 0145 -6

0146 $y=4x+3$ 0147 ② 0148 4 0149 $-\frac{1}{3}<m<1$

0150 ② 0151 $-\frac{1}{2}$ 0152 1 0153 ② 0154 ① 0155 ④

0156 $4\sqrt{5}$ 0157 ⑤ 0158 $\frac{3}{2}$ 0159 ② 0160 ② 0161 ④

0162 9 0163 ⑤ 0164 ④ 0165 -3 0166 2 0167 $\frac{5}{2}$

0168 ② 0169 $y=2x+1$ 0170 ② 0171 ①

0172 $y=\frac{3}{2}x-9$ 0173 ③ 0174 $\frac{22}{3}$ 0175 1 0176 ①

0177 ③ 0178 $24x-10y-143=0$ 0179 ① 0180 ⑤ 0181 2

0182 $y=7x-10$ 0183 ⑤ 0184 -7 0185 $\sqrt{2}$ 0186 5

0187 $\sqrt{5}$ 0188 ③ 0189 4 0190 5 0191 20 0192 ③

0193 $3\sqrt{5}$ 0194 ② 0195 4 0196 $y=-x+6$ 0197 ②

0198 ② 0199 $y=\frac{1}{5}$ 또는 $4x+3y-3=0$

0200 $4x-3y-51=0$ 또는 $4x-3y+29=0$

C단계 0201 $(5, 5)$ 0202 $\frac{x}{4}+\frac{y}{2}=1$ 0203 ① 0204 $\frac{55}{4}$

0205 ⑤ 0206 ① 0207 ⑤ 0208 -5 0209 ④ 0210 ⑤

0211 $\frac{15\sqrt{17}}{17}$ 0212 $x+2y+5=0$ 0213 $3\sqrt{2}$ 0214 $6\sqrt{3}-4$

0215 ③ 0216 -2 0217 ①

03 원의 방정식

43~63쪽

A단계 0218 $x^2+y^2=9$ 0219 $(x-2)^2+(y+1)^2=4$

0220 $(x-3)^2+(y+4)^2=25$ 0221 $(x+3)^2+(y-2)^2=5$

0222 $(x-2)^2+(y-1)^2=18$ 0223 $x^2+(y-1)^2=10$

0224 $(x+5)^2+(y-2)^2=4$ 0225 $(x-4)^2+(y-1)^2=16$

0226 $(x-7)^2+(y+7)^2=49$ 0227 $(0, 5)$, 6

0228 $(-3, 2)$, 1 0229 $\left(-\frac{3}{2}, \frac{1}{2}\right)$, $\sqrt{2}$ 0230 $k<10$

0231 $k<-\sqrt{3}$ 또는 $k>\sqrt{3}$ 0232 $6x-10y-13=0$

0233 $4x+4y+5=0$ 0234 $x^2+y^2+22x-16y+16=0$

0235 한 점에서 만난다.(접한다.) 0236 만나지 않는다.

0237 서로 다른 두 점에서 만난다. 0238 서로 다른 두 점에서 만난다.

0239 한 점에서 만난다.(접한다.) 0240 만나지 않는다.

0241 (1) $-3\sqrt{2}<k<3\sqrt{2}$ (2) $\pm3\sqrt{2}$ (3) $k<-3\sqrt{2}$ 또는 $k>3\sqrt{2}$

0242 $-6<k<2$ 0243 $y=x\pm2$ 0244 $y=-3x\pm10$

0245 $y=-x-3$, $y=-x+5$ 0246 $2x-y=5$

0247 $x-2y=-10$ 0248 $y=-\frac{\sqrt{3}}{3}x+\sqrt{3}$

0249 (1) $x_1x+y_1y=2$ (2) $x_1=-1$, $y_1=1$ 또는 $x_1=1$, $y_1=1$

(3) $-x+y=2$, $x+y=2$

0250 $x-2y=5$, $2x+y=5$

B단계 0251 $(x-3)^2+(y-2)^2=40$ 0252 ① 0253 8
0254 ⑤ 0255 25π 0256 ③ 0257 ③ 0258 14 0259 ②
0260 ② 0261 24 0262 $(x-4)^2+(y-4)^2=32$ 0263 6
0264 ④ 0265 ③ 0266 -25 0267 $(x-1)^2+(y+2)^2=25$
0268 ④ 0269 ④ 0270 2 0271 5 0272 ①
0273 $(x+5)^2+(y+4)^2=16$ 0274 -2 0275 ④ 0276 25π
0277 $x^2+\left(y-\frac{\sqrt{3}}{3}\right)^2=\frac{1}{3}$ 0278 $8\sqrt{2}$ 0279 8
0280 $(x-2)^2+(y-2)^2=4$ 0281 ③ 0282 31 0283 ⑤
0284 (1) $\left(2, \frac{3}{2}\right)$ (2) 최댓값: 7, 최솟값: 3 0285 ④ 0286 121
0287 11 0288 ③ 0289 $2\sqrt{3}$ 0290 ④ 0291 3 0292 2
0293 5 0294 ④ 0295 $\sqrt{3}$ 0296 ③ 0297 ② 0298 $\sqrt{26}\pi$
0299 6 0300 ④ 0301 9 0302 ③ 0303 18 0304 ②
0305 8 0306 5 0307 $\frac{2}{3}\pi+2$ 0308 $\sqrt{11}$ 0309 ①
0310 ⑤ 0311 2 0312 ⑤ 0313 $\frac{7}{2}$ 0314 120 0315 1
0316 ② 0317 7 0318 ② 0319 $\sqrt{5}$ 0320 $\frac{4\sqrt{15}}{5}$
0321 $k<-\frac{\sqrt{2}}{4}$ 또는 $k>\frac{\sqrt{2}}{4}$ 0322 ④ 0323 2 0324 4
0325 ③ 0326 30 0327 14 0328 ② 0329 5π 0330 $\frac{11}{5}$
0331 $y=2x\pm5$ 0332 20 0333 ③ 0334 15 0335 -8
0336 $y=\frac{5}{3}x+\frac{34}{3}$ 0337 -15 0338 ⑤ 0339 ② 0340 ③
0341 ② 0342 $y=\frac{\sqrt{3}}{3}x+\frac{2\sqrt{3}}{3}$, $y=-\frac{\sqrt{3}}{3}x-\frac{2\sqrt{3}}{3}$
0343 (1) $y=2$, $12x-5y-26=0$ (2) $\frac{36}{13}$ (3) $\frac{54}{13}$ 0344 2 0345 ①
0346 3 0347 $\sqrt{33}$, 3
C단계 0348 24 0349 ② 0350 18 0351 ④ 0352 ②
0353 15 0354 15π 0355 ③ 0356 $5\sqrt{2}$ 0357 17 0358 200
0359 150 0360 80 0361 $3x-4y+2=0$ 0362 ④ 0363 7
0364 ④ 0365 ② 0366 $\frac{3\sqrt{5}}{4}$ 0367 ④ 0368 $\frac{10\sqrt{2}}{3}$

04 도형의 이동

65~80쪽

A단계 0369 $(-2, 3)$ 0370 $(2, -14)$
0371 $(4, -6)$ 0372 $(6, 0)$ 0373 $(0, 3)$ 0374 $(7, -16)$
0375 $m=5$, $n=3$ 0376 $x-5y-7=0$
0377 $(x-5)^2+(y-1)^2=9$ 0378 $5x-2y+12=0$
0379 $y=-x^2+4x+15$ 0380 $3x-2y+18=0$
0381 $(x+1)^2+(y+2)^2=4$ 0382 $(2, 3)$ 0383 $(-2, -3)$
0384 $(-2, 3)$ 0385 $(-3, 2)$ 0386 $(3, -2)$
0387 $3x+2y+4=0$ 0388 $3x+2y-4=0$ 0389 $3x-2y-4=0$
0390 $2x-3y-4=0$ 0391 $y=4x-5$ 0392 $y=-2x^2+7x-3$
0393 $y=-x^2+7$ 0394 $x^2+y^2+4x+4y-12=0$
0395 $y=-x^2-x-2$ 0396 $(x-3)^2+(y+1)^2=5$
0397 $y=-\frac{1}{2}x+\frac{3}{2}$ 0398 $(x-2)^2+(y+3)^2=9$
0399 $y=\frac{3}{4}x+\frac{3}{2}$ 0400 $x^2+y^2+4x+2y+4=0$
0401 $(-1, 6)$ 0402 $(4, 3)$
0403 (1) $a=6-p$, $b=-14-q$ (2) $3x+4y+33=0$
0404 (1) $\left(\frac{-4+p}{2}, \frac{3+q}{2}\right)$ (2) $\frac{1}{3}$ (3) $(-1, 4)$

B단계 0405 -6 0406 ② 0407 ④ 0408 $(-6, 11)$
0409 ② 0410 ③ 0411 $2\sqrt{3}$ 0412 ③ 0413 ④ 0414 2
0415 $\frac{1}{3}$ 0416 11 0417 7 0418 ③ 0419 3 0420 ⑤
0421 ⑤ 0422 ② 0423 -20 0424 40 0425 11 0426 $(1, 0)$
0427 ⑤ 0428 ⑤ 0429 제1사분면 0430 1 0431 ②
0432 ① 0433 $(3, 5)$ 0434 ③ 0435 2 0436 ① 0437 ④
0438 -8 0439 ③ 0440 ④ 0441 $2\sqrt{5}-4$ 0442 -16
0443 $4\pi+8$ 0444 ③ 0445 -2 0446 ④ 0447 -1 0448 ④
0449 2 0450 -25 0451 ① 0452 $\frac{5}{4}$ 0453 $0<r<1$
0454 ③ 0455 ① 0456 ① 0457 ⑤ 0458 ② 0459 ⑤
0460 2 0461 ④ 0462 $\frac{5}{2}$ 0463 $(-2, 8)$ 0464 ③
0465 $\frac{12}{5}$ 0466 ② 0467 $\frac{2}{3}$ 0468 $\sqrt{85}$ 0469 ③ 0470 $(0, 3)$
0471 $10\sqrt{2}$ 0472 ②
C단계 0473 14 0474 ① 0475 $16\sqrt{3}$ 0476 $(7, 5)$ 0477 11
0478 2 0479 -8 0480 128 0481 ② 0482 23 0483 $\frac{81}{44}$
0484 $\frac{7}{27}$ 0485 $\frac{\sqrt{10}}{2}$ 0486 ④ 0487 2 0488 ④

05 집합의 뜻과 표현

83~97쪽

A단계 0489 × 0490 ○ 0491 × 0492 ○
0493 1, 2, 3, 4, 6, 12 0494 ∈ 0495 ∉ 0496 ∈ 0497 ∈
0498 {4, 8, 12, 16, …} 0499 {1, 2, 4, 5, 10, 20} 0500 {x|x는 자연수}
0501 {x|x는 100 이하의 10의 양의 배수}
0502 

0503 

0504 무 0505 유
0506 유, 공 0507 무 0508 3 0509 0 0510 3 0511 $Y\subset X$
0512 $X\subset Y$ 0513 $Y\subset X$ 0514 $X\subset Y$ 0515 ∅ 0516 {a}, {b}, {c}
0517 {a, b}, {a, c}, {b, c} 0518 {a, b, c}
0519 ∅, {∅}
0520 ∅, {1}, {2}, {3}, {4}, {1, 2}, {1, 3}, {1, 4}, {2, 3}, {2, 4}, {3, 4}, {1, 2, 3}, {1, 2, 4}, {1, 3, 4}, {2, 3, 4}, {1, 2, 3, 4}
0521 ㄴ, ㄹ 0522 $A=B$ 0523 $A\neq B$ 0524 $A=B$
0525 ∅, {1}, {3}, {9}, {1, 3}, {1, 9}, {3, 9} 0526 32 0527 31
0528 16 0529 8
B단계 0530 ④ 0531 ① 0532 ③ 0533 ㄷ 0534 ③
0535 ④ 0536 5 0537 ③ 0538 ② 0539 30 0540 29
0541 ① 0542 ④ 0543 7 0544 ① 0545 ④ 0546 3
0547 ④ 0548 5 0549 7 0550 ③ 0551 ③ 0552 ③
0553 ④ 0554 ⑤ 0555 ② 0556 3 0557 14 0558 3
0559 ⑤ 0560 ∅, {4}, {7}, {13}, {4, 7}, {4, 13}, {7, 13}, {4, 7, 13}
0561 ② 0562 10 0563 -1 0564 ③ 0565 ② 0566 -4
0567 ① 0568 ③ 0569 6 0570 63 0571 15 0572 ④
0573 64 0574 48 0575 10 0576 ④ 0577 ② 0578 7
0579 19 0580 ① 0581 28 0582 ③ 0583 ③ 0584 4
0585 16 0586 ① 0587 184 0588 ② 0589 ⑤ 0590 160
C단계 0591 -5 0592 ⑤ 0593 20 0594 ⑤ 0595 24
0596 193 0597 56 0598 10 0599 102 0600 ② 0601 ③
0602 80 0603 ② 0604 720 0605 15 0606 144 0607 ④
0608 228

06 집합의 연산

• 99~115쪽

A단계 0609 $\{a, b, c, d, e\}$ 0610 $\{x|x$는 자연수$\}$
0611 $\{3, 6, 9, \cdots, 99\}$ 0612 $\{x, y, z\}$ 0613 $\{2, 3\}$
0614 $\{x|7\le x<12\}$ 0615 $\{2, 3, 4\}$ 0616 ㄱ, ㄷ 0617 $\{2, 4\}$
0618 $\{1, 2, 5\}$ 0619 $\{1, 2, 6, 7, 8\}$ 0620 $\{1, 3, 5, 7\}$
0621 $\{b, c\}$ 0622 $\varnothing$ 0623 $\{3, 5, 6, 7, 8, 9\}$ 0624 $\{1, 4\}$
0625 $\{3, 8, 9\}$ 0626 $\{1, 3, 4, 5, 6, 7, 8, 9\}$ 0627 A
0628 U 0629 $\varnothing$ 0630 A 0631 $\varnothing$ 0632 U 0633 $\{1\}$
0634 $\{7, 8\}$ 0635 $\{2, 4\}$ 0636 $\{2, 4\}$ 0637 (가) 드모르간의 법칙 (나) 결합법칙
0638 $\{4, 8, 10\}$ 0639 $\{1, 2, 3, 5, 6, 7, 9\}$ 0640 2
0641 18 0642 12 0643 26 0644 44 0645 2 0646 40
0647 25 0648 29

B단계 0649 ① 0650 ③ 0651 ④ 0652 ⑤ 0653 ③
0654 8 0655 8 0656 8 0657 ④ 0658 $\{c, d, e\}$
0659 $\{3, 4, 5, 9, 10\}$ 0660 ⑤ 0661 $\{3, 4, 5, 6\}$ 0662 25
0663 8 0664 2 0665 1 0666 5 0667 16 0668 ②
0669 ⑤ 0670 ④ 0671 ⑤ 0672 ③ 0673 ⑤ 0674 ①
0675 4 0676 ① 0677 24 0678 ④ 0679 6 0680 4
0681 14 0682 ④ 0683 $\{2, 4, 6, 7\}$ 0684 ② 0685 ⑤
0686 ⑤ 0687 ㄴ, ㄹ 0688 ② 0689 5 0690 ④ 0691 ②
0692 ④ 0693 ③ 0694 16 0695 6 0696 ④ 0697 24
0698 17 0699 $\{-4, -1, 1, 3\}$ 0700 ③ 0701 $\frac{7}{2}$ 0702 ③
0703 ③ 0704 ② 0705 ③ 0706 ④ 0707 20 0708 ④
0709 55 0710 9 0711 ② 0712 6 0713 21 0714 ②
0715 8 0716 21 0717 ⑤ 0718 12 0719 ③ 0720 ④
0721 22

C단계 0722 2 0723 ④ 0724 224 0725 ③ 0726 23
0727 8 0728 127 0729 22 0730 ⑤ 0731 8 0732 8
0733 ⑤ 0734 ③ 0735 12 0736 ④ 0737 13 0738 ②
0739 58

0774 (1) 필요조건 (2) 필요조건 (3) 필요충분조건 0775 (가) = (나) =
0776 (가) ≥ (나) ≥ 0777 (가) $a-b$ (나) ≥ (다) 0
0778 (가) $2\sqrt{ab}$ (나) > (다) > 0779 풀이 105쪽

0780 풀이 105쪽 0781 2 0782 12 0783 (가) $bx-ay$ (나) $\frac{y}{b}$

B단계 0784 ① 0785 ㄱ, ㄴ, ㄷ 0786 ③ 0787 ④
0788 ④ 0789 ⑤ 0790 ④ 0791 ⑤ 0792 1 0793 4
0794 ③ 0795 1 0796 ㄴ, ㄷ 0797 ② 0798 2, 4, 8, 10, 14, 16
0799 ④ 0800 $5\le k<6$ 0801 ④ 0802 ④ 0803 ④
0804 16 0805 3 0806 4 0807 4 0808 1 0809 ⑤
0810 ④ 0811 $\sqrt{2}$ 0812 5 0813 $1<a<8$ 0814 ⑤
0815 ② 0816 ㄱ, ㄴ 0817 ⑤ 0818 6 0819 5 0820 8
0821 $-1\le a\le 3$ 0822 ⑤ 0823 ② 0824 ⑤ 0825 ④
0826 A, B, C 0827 B, 빵집 0828 ③ 0829 ③
0830 ① 0831 ② 0832 ⑤ 0833 ⑤ 0834 ② 0835 ③
0836 ③ 0837 ④ 0838 4 0839 −2 0840 8 0841 ④
0842 4 0843 ② 0844 ② 0845 8 0846 풀이 113쪽
0847 풀이 113쪽 0848 ③ 0849 풀이 114쪽
0850 풀이 114쪽 0851 (가) $\frac{b}{2}$ (나) 0 0852 ④ 0853 ②
0854 풀이 114쪽 0855 ④ 0856 ④ 0857 ③ 0858 $\sqrt{2}$
0859 ④ 0860 4 0861 9 0862 ③ 0863 ④ 0864 8
0865 ② 0866 24 0867 1 0868 −16 0869 ② 0870 70 m
0871 ③ 0872 60 0873 1 0874 36 0875 ① 0876 ①
0877 10 0878 3 0879 ① 0880 36 0881 ④ 0882 ④
0883 36 0884 4

C단계 0885 ② 0886 ⑤ 0887 17 0888 −9
0889 성민, 진호 0890 ③ 0891 ② 0892 ① 0893 ③
0894 ③ 0895 1 0896 ㄴ, ㄷ 0897 28 0898 ② 0899 4
0900 6

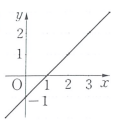
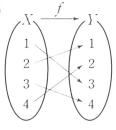
07 명제

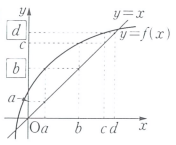
• 117~139쪽

A단계 0740 참인 명제: ㄷ, ㅂ, 거짓인 명제: ㄱ, ㄹ 0741 ㄱ, ㄹ
0742 $\{-4, 1\}$ 0743 $\{2, 3, 5, 7\}$
0744 $\sqrt{4}$는 무리수가 아니다. (참) 0745 1은 합성수이거나 소수이다. (거짓)
0746 $x\ne1$이고 $x\ne2$ 0747 $x<-2$ 또는 $x\ge3$ 0748 $\{3, 4, 6, 7, 8, 9\}$
0749 $\{1, 3, 5, 7, 9\}$ 0750 $\{1, 2, 4, 5, 6, 8, 10\}$ 0751 $\{2, 10\}$
0752 $\{2, 3, 4, 6, 7, 8, 9, 10\}$ 0753 $\{1, 5\}$
0754 가정: 18의 약수이다., 결론: 9의 약수이다.
0755 가정: $ab=0$이다., 결론: $a=0$ 또는 $b=0$이다.
0756 거짓 0757 참 0758 거짓 0759 참 0760 참 0761 거짓
0762 어떤 실수 x에 대하여 $x^2<0$이다. (거짓)
0763 모든 실수 x에 대하여 $x^2+1\ne0$이다. (참)
0764 역: $x=1$이면 $x^2=1$이다., 대우: $x\ne1$이면 $x^2\ne1$이다.
0765 역: $x=0$이고 $y=0$이면 $x^2+y^2=0$이다.
대우: $x\ne0$ 또는 $y\ne0$이면 $x^2+y^2\ne0$이다.
0766 역: $a>0$ 또는 $b>0$이면 $a+b>0$이다.
대우: $a\le0$이고 $b\le0$이면 $a+b\le0$이다.
0767 ㄷ
0768 (1) x, y가 모두 유리수이면 $x+y$는 유리수이다. (2) 참 (3) 참
0769 ㄴ, ㄷ 0770 충분조건 0771 필요충분조건
0772 필요조건 0773 충분조건

08 함수

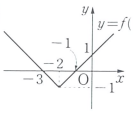
• 143~165쪽

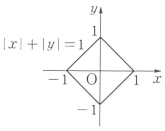
A단계 0901 함수가 아니다.
0902 함수이다., 정의역: {1, 3, 5, 7}, 공역: {1, 2}, 치역: {1, 2}
0903 함수이다., 정의역: $\{-1, 0, 1\}$, 공역: {1, 3, 5, 7}, 치역: {1, 5, 7}
0904 함수가 아니다.
0905 정의역: 실수 전체의 집합, 치역: 실수 전체의 집합
0906 정의역: 실수 전체의 집합, 치역: $\{y|y\le4\}$
0907 정의역: $\{x|x\ne0$인 실수$\}$, 치역: $\{y|y\ne0$인 실수$\}$
0908 서로 같은 함수이다. 0909 서로 같은 함수가 아니다.
0910 
0911 
0912 ○ 0913 ×
0914 ㄴ, ㄹ 0915 ㄴ, ㄹ 0916 ㄹ 0917 ㄱ 0918 ㄱ, ㄷ 0919 ㄱ, ㄷ
0920 ㄷ 0921 ㄴ 0922 6 0923 6 0924 d 0925 b
0926 $(g\circ f)(x)=9x^2-6x+1$ 0927 $(f\circ g)(x)=3x^2-1$
0928 $(f\circ f)(x)=9x-4$ 0929 $(g\circ g)(x)=x^4$ 0930 풀이 125쪽
0931 (1)  (2) 5 (3) 1 0932 1 0933 −1
0934 $y=\frac{1}{2}x-\frac{3}{2}$ 0935 $y=2x+\frac{1}{2}$ 0936 3 0937 1

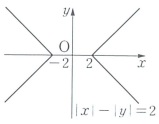
0938 5 0939 4 0940  0941 b 0942 c

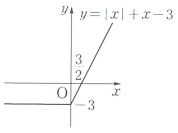
0943 $(-2, -2)$ 0944 $(1, 1)$ 0945 $(3, 3)$

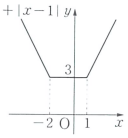
0946 (1) $f(x)=x+1$ (2) $f(x)=-x-3$

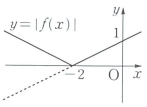
(3)  , 치역: $\{y|y\geq -1\}$

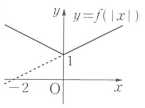
0947 

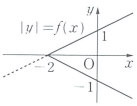
0948 

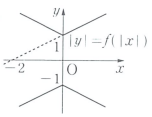
0949 

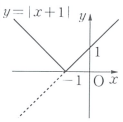
0950 

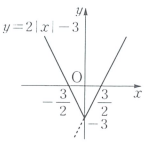
0951 

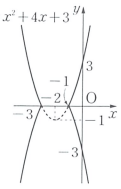
0952 

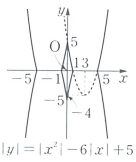
0953 

0954 

0955 

0956 

0957 

0958 

B 단계

0959 ③ 0960 ㄱ, ㄹ 0961 ⑤ 0962 6 0963 ④

0964 8 0965 $\begin{pmatrix}0&0&1\\0&1&0\\0&1&0\end{pmatrix}$ 0966 384 0967 ③ 0968 17

0969 ⑤ 0970 -2 0971 3 0972 20 0973 $1\leq a\leq \frac{5}{2}$

0974 ① 0975 ③ 0976 ② 0977 $\{2, 3\}$ 0978 $\{1\}$, $\{3\}$, $\{1, 3\}$

0979 ㄴ, ㅁ 0980 5 0981 7 0982 ④ 0983 ④ 0984 $\frac{29}{4}$

0985 $-2<a<2$ 0986 8 0987 3 0988 2 0989 1

0990 12 0991 10 0992 6 0993 ⑤ 0994 ④ 0995 ③

0996 9 0997 ① 0998 ② 0999 -8 1000 -2 1001 ⑤

1002 ② 1003 ④ 1004 $(-3, -3)$ 1005 -12 1006 ⑤

1007 3 1008 5 1009 $\sqrt{10}$ 1010 ③ 1011 5

1012 (1) $h(x)=-\frac{1}{3}x+1$ (2) $k(x)=-\frac{1}{3}x+\frac{1}{3}$ 1013 ② 1014 4

1015 ③ 1016 $f_n(x)=\begin{cases}-x+2 & (n\text{은 홀수})\\ x & (n\text{은 짝수})\end{cases}$ 1017 ③ 1018 ⑤

1019 ④ 1020 1 1021 16 1022 ③ 1023 4 1024 $\frac{11}{4}$

1025 13 1026 ③ 1027 3 1028 2 1029 ② 1030 ③

1031 8 1032 5 1033 $k<-1$ 또는 $k>1$ 1034 ⑤ 1035 ②

1036 $h^{-1}(x)=-\frac{1}{2}x-\frac{11}{2}$ 1037 ② 1038 $\frac{5}{2}$ 1039 ⑤

1040 ⑤ 1041 3 1042 3 1043 (1) $\frac{28}{3}$ (2) -25 1044 -6

1045 ① 1046 3 1047 ② 1048 ⑤ 1049 7 1050 s

1051 ③ 1052 2 1053 ② 1054 $2\sqrt{2}$ 1055 $2\leq a<\frac{9}{4}$

1056 243 1057 ④ 1058 ㄴ, ㄱ 1059 ② 1060 -25 1061 2

1062 ②

C 단계 1063 7 1064 ⑤ 1065 88 1066 2 1067 ⑤

1068 ③ 1069 12 1070 40 1071 $-\frac{7}{4}<a<\frac{9}{4}$ 1072 $\frac{1}{2}$

1073 ㄴ, ㄷ 1074 ③ 1075 16 1076 $\frac{361}{16}$ 1077 ③ 1078 6

1079 $\frac{1}{2}<m<\frac{7}{5}$

09 유리식과 유리함수

167~183쪽

A 단계 1080 ㄴ, ㄷ 1081 ㄱ, ㄹ, ㅁ

1082 $\frac{3b^2x}{6a^2bx^2y}$, $\frac{2ay}{6a^2bx^2y}$

1083 $\frac{(x+1)(x+2)}{(x-1)(x-2)(x+2)}$, $\frac{(x-1)^2}{(x-1)(x-2)(x+2)}$ 1084 $\frac{3z}{4y}$

1085 $x+2$ 1086 $\frac{-2x-1}{(x+2)(x+5)}$ 1087 $\frac{x+6}{x+3}$ 1088 $\frac{2x+1}{(x+2)(x-2)}$

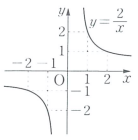
1089 $\frac{(x+2)(x-1)}{(x+3)(2x-1)}$ 1090 $\frac{4x-7}{(x-3)(x-2)}$ 1091 $\frac{3x-5}{(x+1)(x-1)}$

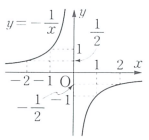
1092 $\frac{2}{(x+1)(x+3)}$ 1093 $\frac{3}{x(x+3)}$ 1094 $\frac{x}{x-1}$ 1095 $x-1$

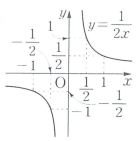
1096 $\frac{1}{5}$ 1097 $\frac{13}{25}$ 1098 ㄱ, ㄹ 1099 ㄴ, ㄷ, ㅁ

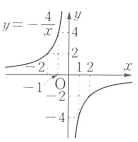
1100 $\left\{x\middle|x\neq -\frac{3}{2}\text{인 실수}\right\}$ 1101 $\{x|x\neq 5\text{인 실수}\}$

1102 $\{x|x\neq \pm 2\text{인 실수}\}$ 1103 실수 전체의 집합

1104 

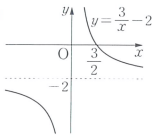
1105 

1106 

1107 

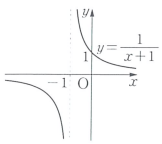
1108 $y=\frac{1}{x-1}+2$

1109 $y=-\frac{2}{x+2}-1$

1110 

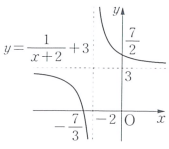
정의역: $\{x|x\neq 0\text{인 실수}\}$

치역: $\{y|y\neq -2\text{인 실수}\}$

1111 

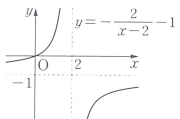
정의역: $\{x|x\neq -1\text{인 실수}\}$

치역: $\{y|y\neq 0\text{인 실수}\}$

1112 

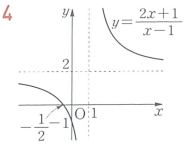
정의역: $\{x|x\neq -2\text{인 실수}\}$

치역: $\{y|y\neq 3\text{인 실수}\}$

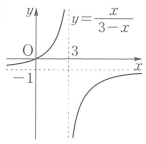
1113 

정의역: $\{x|x\neq 2\text{인 실수}\}$

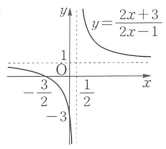
치역: $\{y|y\neq -1\text{인 실수}\}$

1114 

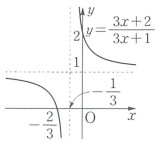
$x=1$, $y=2$

1115 

$x=3$, $y=-1$

1116 

$x=\frac{1}{2}$, $y=1$

1117 

$x=-\frac{1}{3}$, $y=1$

B 단계

1118 $\frac{1}{x^2-x+1}$ 1119 ③

1120 $\frac{6x^2+6x-6}{x(x+3)(x+1)(x-2)}$ 1121 $\frac{a}{a+1}$ 1122 $\frac{5}{6}$ 1123 $\frac{x-1}{x+2}$

1124 ③ 1125 1 1126 64 1127 ② 1128 ①

1129 $\frac{11x-9}{x(x+1)(3x-1)}$ 1130 -1 1131 $\frac{-2x}{x^4+x^2+1}$ 1132 18

1133 6 1134 ③ 1135 6 1136 $\frac{49}{99}$ 1137 $\frac{x-1}{x}$ 1138 ④

1139 4 1140 ④ 1141 ③ 1142 $\frac{1}{6}$ 1143 1 1144 $\frac{1}{3}$

1145 7 1146 ③ 1147 4 1148 4 1149 5 1150 -7

1151 3 1152 ④ 1153 ② 1154 24 1155 2 1156 3

1157 2 1158 ② 1159 $-\frac{3}{2}$ 1160 ⑤ 1161 제1사분면

1162 $a\le-2$ 1163 $k>4$ 1164 8 1165 ② 1166 ③

1167 ④ 1168 ③ 1169 $a<b<c<d$ 1170 $\frac{5}{3}$ 1171 ⑤

1172 -1 1173 $-\frac{7}{8}$ 1174 12 1175 -1 1176 $a<0$ 1177 ③

1178 -3 1179 ③ 1180 $\sqrt{10}$ 1181 18 1182 ④ 1183 $\frac{7}{8}$

1184 ③ 1185 -9 1186 0 1187 ① 1188 -24 1189 ③

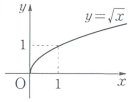
1190 -8 1191 ④ 1192 $\frac{5}{3}$ 1193 -1

C 단계

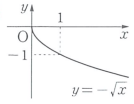
1194 $\frac{11}{6}$ 1195 50 1196 ⑤ 1197 88 1198 ④

1199 102 1200 24 1201 11 1202 ③ 1203 -1 1204 1

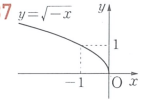
1205 192 1206 ⑤ 1207 ⑤ 1208 ③ 1209 ⑤

1235 

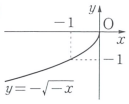
정의역: $\{x|x\ge0\}$
치역: $\{y|y\ge0\}$

1236 

정의역: $\{x|x\ge0\}$
치역: $\{y|y\le0\}$

1237 

정의역: $\{x|x\le0\}$
치역: $\{y|y\ge0\}$

1238 

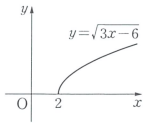
정의역: $\{x|x\le0\}$
치역: $\{y|y\le0\}$

1239 $y=-\sqrt{-3x}$ 1240 $y=\sqrt{3x}$

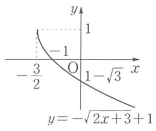
1241 $y=-\sqrt{3x}$ 1242 $y=\sqrt{x-3}-2$ 1243 $y=\sqrt{-4x-4}+5$

1244 $y=-\sqrt{7x+42}-4$ 1245 $y=\sqrt{4(x+2)}-6$

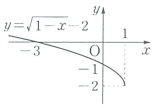
1246 $y=-\sqrt{-5\left(x-\frac{3}{5}\right)}+1$ 1247 $a=2$, $b=2$

1248 

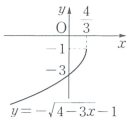
정의역: $\{x|x\ge2\}$
치역: $\{y|y\ge0\}$

1249 

정의역: $\left\{x\middle|x\ge-\frac{3}{2}\right\}$
치역: $\{y|y\le1\}$

1250 

정의역: $\{x|x\le1\}$
치역: $\{y|y\ge-2\}$

1251 

정의역: $\left\{x\middle|x\le\frac{4}{3}\right\}$
치역: $\{y|y\le-1\}$

B 단계

1252 $x\le-\frac{4}{3}$ 또는 $x\ge\frac{1}{2}$ 1253 12 1254 $-x$

1255 ⑤ 1256 ③ 1257 ⑤ 1258 $3a$ 1259 ③ 1260 ②

1261 ④ 1262 $2\sqrt{2}$ 1263 ④ 1264 $\frac{\sqrt{5}}{5}$ 1265 ③ 1266 82

1267 ③ 1268 2 1269 $2\sqrt{3}$ 1270 ④ 1271 ③ 1272 ④

1273 ④ 1274 정의역: $\left\{x\middle|x\le\frac{4}{5}\right\}$, 치역: $\{y|y\ge-1\}$ 1275 ②

1276 9 1277 ④ 1278 -5 1279 3 1280 ④ 1281 3

1282 ② 1283 -4 1284 ⑤ 1285 2 1286 ① 1287 7

1288 $a<0$, $b<0$, $c<0$ 1289 ① 1290 ④ 1291 ④ 1292 ①

1293 ③ 1294 2 1295 ③ 1296 ③ 1297 8 1298 20

1299 ⑤ 1300 1 1301 ④ 1302 4 1303 ② 1304 -9

1305 ① 1306 -1 1307 ④ 1308 ① 1309 2 1310 ③

1311 ⑤ 1312 2 1313 $\{y|2<y<3\}$

C 단계

1314 $bc-ad-1$ 1315 45 1316 2 1317 ⑤

1318 ① 1319 ③ 1320 $\frac{9}{8}$ 1321 7 1322 ② 1323 ③

10 무리식과 무리함수

185~199쪽

A 단계

1210 $x\ge-2$ 1211 $x\ge1$ 1212 $2\le x<3$

1213 $x\ge0$ 1214 $-1\le x<5$ 1215 $2a-3b$ 1216 1

1217 $x-10$ 1218 -2 1219 2 1220 $\sqrt{x+1}-\sqrt{x}$

1221 $\sqrt{x+2}+\sqrt{x-2}$ 1222 $\frac{a-2\sqrt{ab}+b}{a-b}$

1223 $-2x+1-2\sqrt{x(x-1)}$ 1224 $2x^2-1+2x\sqrt{x^2-1}$ 1225 $\frac{2\sqrt{a}}{a-b}$

1226 $\frac{x+3}{x-3}$ 1227 $\frac{-2(x+\sqrt{x+1})}{x}$ 1228 $\frac{-\sqrt{x}+\sqrt{x+6}}{2}$

1229 $2(2+\sqrt{2})$ 1230 ㄱ, ㄷ, ㅁ, ㅂ 1231 $\{x|x\ge-1\}$

1232 $\{x|x\le0\}$ 1233 $\left\{x\middle|x\ge\frac{3}{2}\right\}$ 1234 $\{x|-2\le x\le2\}$